NIRNet: Noise Incentive Robust Network in Remote Sensing Object Detection Under Cloud Corruption

Peng Zhang, Gong Cheng, *Senior Member, IEEE*, Chunbo Lang, Xingxing Xie, and Junwei Han, *Fellow, IEEE*

***Abstract*— Within remote sensing images, complex atmospheric environments commonly bring about distinct variations in imaging visibility and ambient occlusions, significantly transforming the appearance of objects. Nevertheless, modern detectors generally struggle to maintain promising accuracy when encountering realistic scenarios. Devoting to alleviating the issues, we develop a noise incentive robust network (NIRNet) for remote sensing object detection under cloud corruption without relying on hazy images for training. The proposed NIRNet preserves discriminative representations and calibrates them using an incentive mechanism. First, we design a noise perception module (NPM) to deal with diverse cloud corruption types, which generates pointwise calibration weights dependent on the perceived discrepancy between objects and environmental noise. Second, aiming to detect difficult-to-discern objects thoroughly, a dual-path incentive calibration (DPIC) strategy is proposed to combine intensity and stability features weighted by NPM. Profiting from its universal design, the DPIC could be treated as a plug-and-play module for existing detectors, enhancing robustness against adverse weather. To evaluate the reliability of aerial detectors under intricate cloud corruptions, we present an elaborate Hazy-DIOR dataset, which contains numerous images with different cloud conditions and severity levels. Finally, extensive experiments on the Hazy-DIOR and DOTA-Cloud datasets simultaneously demonstrate the robustness of NIRNet, which especially achieves state-of-the-art accuracy and gets 2.16% mean average precision (mAP) and 2.56% relative performance under corruption (rPC) improvements on the Hazy-DIOR compared with solid Oriented R-CNN detector. The code is available at: https://github.com/zhangpeng2001/nirnet**

***Index Terms*— Cloud corruption, Hazy-DIOR dataset, remote sensing images, robust object detection.**

I. Introduction

DEEP learning-based computer vision technology [1], [2], [3], [4], [5], [6], [7], [8], [9] has witnessed considerable breakthroughs over the past decade. Extensive pioneering works in object detection attempt to develop a faster and stronger detector [10], [11], [12], [13], [14], [15]. Meanwhile, numerous essential components, specially designed for the orientation arbitrariness and scale diversity in remote sensing images, promote the application of aerial detectors [16], [17], [18], [19]. Thanks to the excellent representation learning ability, existing approaches excel at categorizing and positioning most objects in clean images with the same distribution as ones from the training set. However, their accuracy often decreases sharply in realistic environments due to the common influences of image distortions [20], [21], [22] or adverse weather conditions [23], [24], [25]. Distinct from everyday scenarios, satellite imagery and aerial photography frequently suffer from intricate cloud occlusion, where the reduced atmospheric visibility leads to a significant degradation in detection accuracy [26].

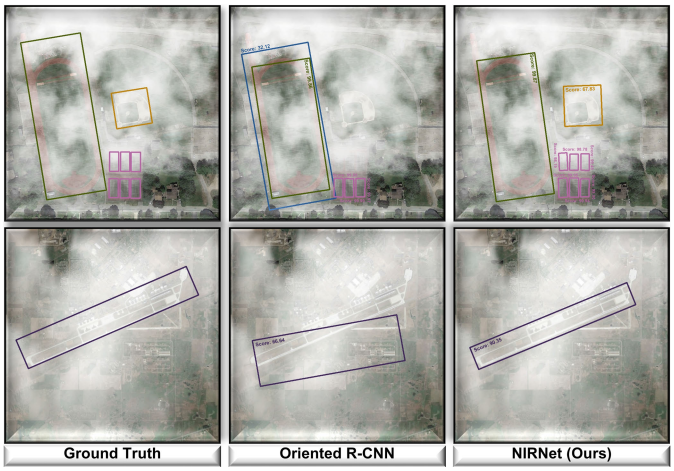


Fig. 1. Visualization of rotated object detection on remote sensing images under cloud corruption. The *ground truth* from Hazy-DIOR is presented in the first column, while the second and third columns show the results of Oriented R-CNN and our proposed NIRNet. The colored bounding boxes indicate the detected objects in each scenario (● STA, ● BF, ● GTF, ● TC, and ● APO). The famous aerial detector (Oriented R-CNN [17]) fails to detect the BF and TC s while incorrectly identifying an STA in the first row. Meanwhile, it imprecisely predicts the bounding box of an APO in the second row, especially angle regression. However, our NIRNet could mitigate these inaccurate problems and achieve robust remote sensing object detection against cloud corruption.

Observational data indicate that the cloud fraction over land is approximately 55%, showing a distinct seasonal cycle; while over the ocean, it is about 72% with less pronounced seasonal variation [27]. Meanwhile, atmospheric corruption constantly changes the appearance of instances, whether due to reduced visibility or ambient occlusion. Consequently, persistent cloud cover challenges the widespread application of aerial detectors. The models trained using clean images conventionally create tremendous performance gaps when encountering cloud occlusion. As evidenced in Fig. 1, the famous baseline

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The authors are with the School of Automation, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: gcheng@nwpu.edu.cn).

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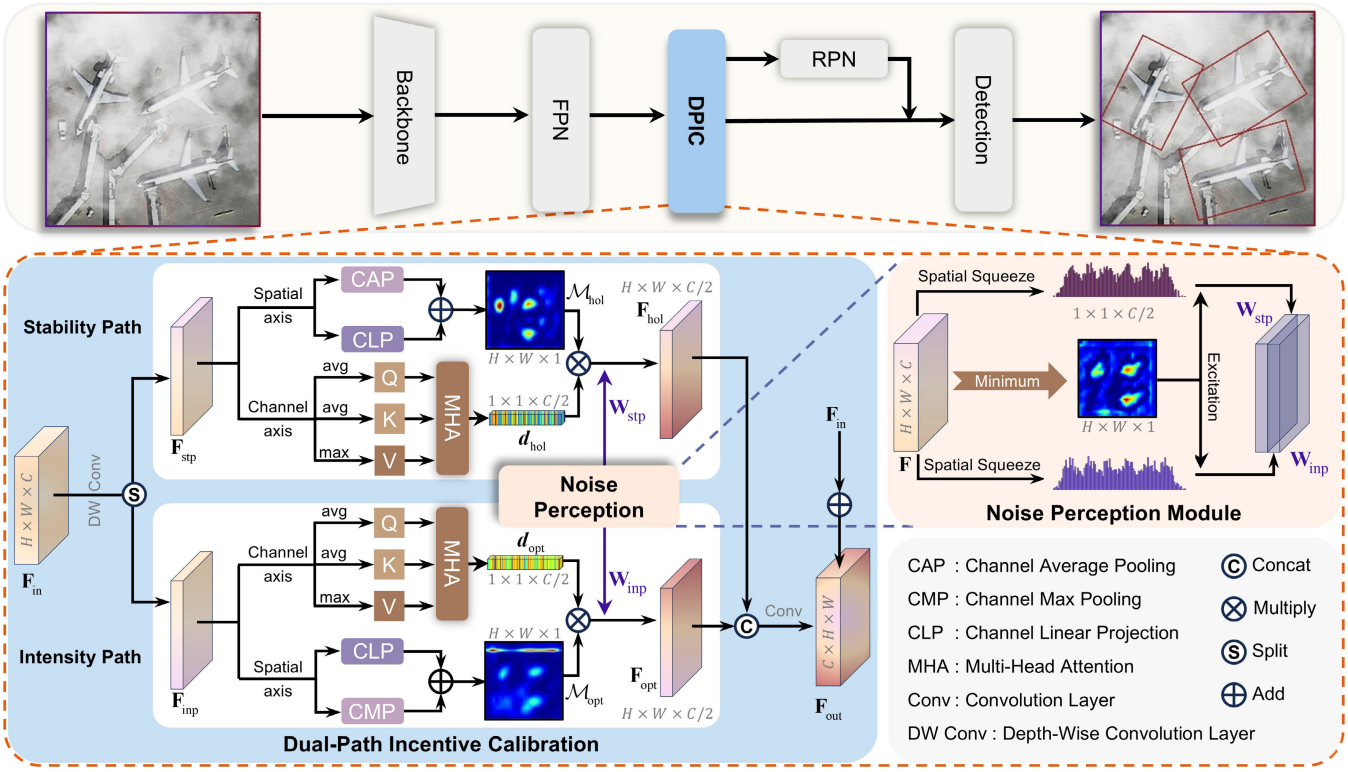


Fig. 2. Overall architecture of the proposed NIRNet, illustrated with the inference pipeline. It detects objects in hazy remote sensing images by incorporating DPIC into a detection network, exemplified by Oriented R-CNN [17]. The DPIC employs a parallel strategy, with the stability path concentrating on holistic information and the intensity path on optimal features. The NPM is essential for generating incentive weights to emphasize object areas. When the calibration mechanism is applied without NPM, the method is named FCM.

(Oriented R-CNN [17]) in remote sensing object detection struggles to accurately recognize objects in hazy images, which results in misclassification, imprecise regression, and missing detection.

To cope with the detrimental effect of adverse weather corruption, numerous domain adaptation (DA) methods have been proposed [28], [29], [30]. In particular, DA endeavors to narrow the disparity between clean and corrupted images by aligning their data distributions and learning domain-invariant feature representation from the source domain [31], [32], [33]. However, DA approaches heavily rely on an essential assumption that target-domain images are available, which are expensive and time-consuming to obtain or may even be unknown before deploying models [34]. Meanwhile, several recent works concentrate on recovering images before detection or segmentation tasks, where the introduction of additional networks reduces their efficiency for real-time implementation [35], [36], [37], [38].

The indistinct boundaries between interested instances and backgrounds deriving from the cloud disturbance produce the challenge of detecting objects. Drawing inspiration from [39], [40], we design a noise-perception feature calibration mechanism and construct the noise incentive robust network (NIRNet), which mitigates the performance degradation from cloud corruption. Specifically, we develop a noise perception module (NPM) to generate calibration excitations by distinguishing the discrepancy between objects and background noise. Operating independently of specific input, NPM identifies potential objects from the minimal responses to deal with different cloud corruption types. In addition, with the goal of improving the discriminative capacity of feature representation, a dual-path incentive calibration (DPIC) accepts the corrupted feature as input to generate intensity and stability features that are further emphasized with excitations provided by NPM. As exhibited in Fig. 2, we construct NIRNet by integrating DPIC into any existing detector, e.g., Oriented R-CNN [17]. Despite being trained without hazy images, it achieves preferable accuracy in remote sensing object detection against cloud corruption. Meanwhile, the operations decoupled into spatial and channel views of the calibration mechanism maintain the efficiency of DPIC. In addition, benefiting from its modularized architecture, DPIC is considered a plug-and-play component for the most popular detectors, achieving robust object detection in remote sensing images under cloud corruption.

Aiming at the problem of insufficient datasets to evaluate the robustness of aerial object detectors, we create a cloud-corrupted remote sensing dataset named Hazy-DIOR, which covers a wide range of cloud corruption types. Specifically, we utilize the raw data from the DIOR-R testing set to synthesize different hazy remote sensing images, as obtaining a sufficient number of geoscience images with interested objects is costly and time-consuming. It provides a comprehensive benchmark for assessing performance degradation in remote sensing object detection under high-altitude cloud occlusions.

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In brief, our contributions are summarized as follows.

1) We construct an NIRNet for remote sensing object detection against cloud occlusions without relying on hazy images for training.
2) We design an NPM to identify potential objects by distinguishing the discrepancy between minimal responses of objects and ambient noise.
3) We develop an incentive calibration mechanism DPIC to emphasize intensity and stability features with pointwise weights for discriminative representation.
4) We establish a new benchmark for robust remote sensing object detection under cloud corruption named Hazy-DIOR, which contains diverse hazy conditions.

II. Related Works

A. Object Detection in Remote Sensing Images

Mainstream detection algorithms fall into two types: one-stage detectors [11], [41] and two-stage detectors [10], [42]. Both types extract the pyramid features using a standard backbone [1], [43], followed by a multiscale feature fusion algorithm [44], [45]. In most cases, one-stage detectors predict the location and category of any interested object utilizing a unified architecture, while two-stage models rely on region-based representations. Plentiful studies contribute to feasible solutions for recognizing objects with arbitrary orientations in remote sensing object detection [46], [47], [48]. Ding et al. [16] introduced a region of interests (RoI) Transformer to rotate horizontal proposals for extracting rotation-invariant features. Oriented R-CNN [17] presents an oriented region proposal network (RPN) for generating high-quality proposals with minimal computational cost. Single-shot align network (S2ANet) [18] mitigates the misalignment between horizontal features and oriented instances. Refined rotation retinanet (R3Det) [19] exhibits a coarse-to-fine regression approach, achieving an end-to-end one-stage detector. Meanwhile, many efforts [49], [50], [51], [52], [53], [54], [55] in the remote sensing community concentrate on developing a more suitable and powerful aerial detector. Beyond that, several works [56], [57], [58] are devoted to promoting detection performance in autonomous aerial vehicle images, and other works aim to solve the detection problem from different image sources, such as thermal infrared images [59], [60] and synthetic aperture radar images [61], [62].

With regard to the severe scale variations in remote sensing, several researchers resolve small object detection in large-scale images [63], [64], [65]. Yuan et al. [63] designed a coarse-to-fine RPN to generate high-quality candidates for small objects. Xu et al. [64] proposed a dynamic optimizing algorithm in all predefined bounding boxes, label assignment, and coordinate representation to alleviate the mismatch phenomenon for small objects. Moreover, several researchers have explored efficient feature extraction networks for aerial detectors [66], [67], [68]. Even so, there are still obstacles in practical scenarios. Detectors often struggle with robustness when attempting to detect challenging objects in hazy remote sensing images. Consequently, we aim to explore the generalization capacity of detectors to mitigate the application barriers posed by cloud corruption.

B. Object Detection in Adverse Weather Conditions

Standard detectors [10], [11], [12], [13], [14] are optimized for ideal tasks (invariant classification and static regression), which results in subpar performance when encountering real-world scenarios [23]. However, massive difficult-to-discern objects reduce the detection performance in adverse weather conditions [38]. To alleviate the problem, several DA algorithms leverage labeled source-domain images and unlabeled target-domain images to train the network through knowledge transformation from source domain to target domain [24], [29]. Most of these methods heavily depend on multidomain data during training, while obtaining abundant images against realistic adverse weather is challenging and expensive, particularly remote sensing images under cloud occlusion.

Several restoration algorithms recover high-quality images before object detection, which are time-consuming due to the introduction of another complex network and still suffer from inaccurate predictions. Huang et al. [38] sought to increase the accuracy under inclement conditions by implanting visibility enhancement modules into the detector. Li et al. [36] combined the restoration and detection components, synthesizing a two-branch framework to simultaneously detect objects on a hazy image and its corresponding restored one to decrease negative impacts on object detection. Sun et al. [35] designed a targeted adversarial attack after the restoration to strengthen the detection performance. However, this strategy is computationally expensive and inefficient for real-time detection. In remote sensing imagery, many studies are devoted to addressing the challenge of dehazing [69], [70], but fewer researchers focus on enhancing the robustness of object detectors under cloud occlusion. Compared to the restoration framework, we propose an end-to-end NIRNet for hazy aerial images, which does not rely on target domain data. In the training phase, discriminative features of difficult-to-discern objects are focused spatially and calibrated via a noise incentive mechanism. In the testing phase, NIRNet identifies the discrepancies between objects and ambient noise to cope with diverse cloud occlusions.

III. Our Method

This section presents a comprehensive overview of robust remote sensing object detection under cloud corruption. We begin by outlining the objectives of our strategy and detailing the pipeline of the NIRNet. Next, we describe the overall architecture of the DPIC strategy. Finally, we present the structure of the NPM.

A. Pipeline of NIRNet

Existing aerial detectors [16], [17], [18], [19] often prove inadequate to recognize challenging objects in hazy remote sensing images, which impedes their widespread implementation in real-world scenarios [71]. Convolution-based models, as the mainstream detection algorithms, accept the feature hierarchy to forecast category labels and their corresponding

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locations of all potential objects [10], [17]. The robustness of these detectors depends heavily on semantic features employed in the detection heads (classification and regression heads) [44]. Therefore, we propose the DPIC strategy to strengthen feature distinctiveness for these heads, enabling detectors to recognize difficult-to-discern objects accurately.

As depicted in Fig. 2, DPIC consists of two essential subcomponents: NPM and feature calibration module (FCM). The NPM exploits pooling operations in a novel fashion, aiming to locate potential regions based on the analysis of minimal responses. By decoupling the input features, the FCM facilitates the concurrent extraction of the holistic and optimal representations. In addition, in FCM, the attention mechanism captures global dependencies to generate comprehensive feature descriptors, while the auxiliary linear projection enhances the learnability of the feature masks.

When a remote sensing image $\mathbf{X} \in \mathbb{R}^{h\times w\times c}$ is provided as input (where $h \times w$ represents image size and c typically equals 3 for color channels), the backbone network downsamples the image employing convolutional layers to generate a pyramid hierarchy $\mathbf{P}_i \in \mathbb{R}^{H_i\times W_i\times 256}$ (i indicates the index of feature level ranging from 3 to 7 and $H_i \times W_i$ represents to its size). DPIC accepts a single-feature map from the pyramid (written as $\mathbf{F}_{\text{in}}$) and generates a discriminative feature ($\mathbf{F}_{\text{out}}$), which is then used to predict oriented proposals and detection results. Notably, DPIC shares the parameters across different levels on the feature pyramid hierarchy. During the training phase, classification and regression losses are prerequisites for the network to update parameters.

B. Dual-Path Incentive Calibration

To improve the robustness of detectors, we design a DPIC, which concurrently focuses on the optimal and holistic representations of potential objects. Given an input $\mathbf{F}_{\text{in}} \in \mathbb{R}^{H\times W\times C}$ from the pyramid hierarchy, DPIC first predicts a spatial sensitive feature $\mathbf{F}$ and decouples it into two subfeatures with the same dimension. Its mathematical formula is defined as follows:

$$\mathbf{F}_{\text{stp}},\ \mathbf{F}_{\text{inp}} = \mathcal{S}(\Psi(\mathbf{F}_{\text{in}}, \theta)) \tag{1}$$

where $\mathcal{S}(\cdot)$ denotes a split operation along the channel axis. $\Psi(\cdot)$ represents a depthwise convolution layer followed by normalization and activation functions, and θ is their inner parameters. Furthermore, the activated feature $\mathbf{F}$ is divided into two subfeatures: $\mathbf{F}_{\text{stp}}$ for stability-path feature calibration and $\mathbf{F}_{\text{inp}}$ for intensity-path feature calibration.

The intensity path calibration branch focuses on the maximum values to directly distinguish significant objects, while the stability branch aggregates the mean values to address ambiguous objects thoroughly. Furthermore, NPM, introduced in Section III-C, calibrates the obtained features as follows:

$$\begin{cases} \mathbf{F}_{\text{hol}} = \mathcal{SPC}(\mathbf{F}_{\text{stp}}, \theta_{\text{stp}}) * \mathbf{W}_{\text{stp}} \\ \mathbf{F}_{\text{opt}} = \mathcal{IPC}(\mathbf{F}_{\text{inp}}, \theta_{\text{inp}}) * \mathbf{W}_{\text{inp}} \end{cases} \tag{2}$$

where $*$ is the elementwise multiplication. Parameters θ_{stp} within stability path calibration mechanism $\mathcal{SPC}(\cdot)$ are optimized to acquire holistic information from the spatially sensitive feature $\mathbf{F}_{\text{stp}}$. Similarly, parameters θ_{inp} within the intensity path calibration mechanism $\mathcal{IPC}(\cdot)$ are optimized to capture optimal information from the spatially sensitive feature $\mathbf{F}_{\text{inp}}$. Ultimately, the extracted features are element-by-element calibrated by NPM: $\mathbf{W}_{\text{stp}}$ for the stability path and $\mathbf{W}_{\text{inp}}$ for the intensity path branches.

After parallel calibration, DPIC integrates two discriminative features to generate the final output with a convolutional module as expressed in the following formula:

$$\mathbf{F}_{\text{out}} = \Phi\big(\mathcal{C}\big(\mathbf{F}_{\text{hol}},\ \mathbf{F}_{\text{opt}}\big)\big) + \mathbf{F}_{\text{in}} \tag{3}$$

where $\mathcal{C}(\cdot)$ denotes a concatenation operation along the channel axis and $\Phi(\cdot)$ is a convolutional layer followed by normalization and activation functions. The DPIC without noise incentive mechanism is named FCM in Sections III-C and IV. Therefore, the incentive calibration mechanism serves as an integration of two separate but synergistic components. In addition, both feature calibration processes are decoupled along the spatial and channel axes, simultaneously generating a descriptor and its corresponding mask. The entire process can be summarized as follows:

$$\mathbf{Y}_i = \mathcal{M}_i \otimes \boldsymbol{d}_i \tag{4}$$

where $\otimes$ represents the matrix multiplication. The calibration mechanism multiplies mask $\mathcal{M}_i$ and descriptor $\boldsymbol{d}_i$ to generate the output $\mathbf{Y}_i$ in each path. The optimal mask $\mathcal{M}_{\text{opt}}$ and its corresponding descriptor $\boldsymbol{d}_{\text{opt}}$ are the products of intensity path feature calibration. Similarly, the holistic mask $\mathcal{M}_{\text{hol}}$ and its corresponding descriptor $\boldsymbol{d}_{\text{hol}}$ are the outcomes of stability path feature calibration.

In the intensity path branch, channel max pooling (CMP) obtains the local but optimal representation, defining the significant signals for each point in the feature map. To address the limitations of single pooling, a learned auxiliary coefficient adjusts the value at each point. Ultimately, the optimal mask $\mathcal{M}_{\text{opt}}$ is defined as follows:

$$\mathcal{M}_{\text{opt}}(i, j) = \max_k(u(i, j, k)) + \psi\big(\mathbf{F}_{\text{inp}}(i, j)\big) \tag{5}$$

where $u(i, j, k) \in \mathbf{F}_{\text{inp}}$ and $\psi(\cdot)$ denotes a linear function with an input dimension of $C = 128$ and an output dimension of 1. Conversely, channel average pooling (CAP) utilizes global information to capture potential signals in the stability path branch. The holistic mask $\mathcal{M}_{\text{hol}}$ is similarly defined as follows:

$$\mathcal{M}_{\text{hol}}(i, j) = \frac{1}{C}\sum_{k=0}^{C-1} u(i, j, k) + \varphi\big(\mathbf{F}_{\text{stp}}(i, j)\big) \tag{6}$$

where $u(i, j, k) \in \mathbf{F}_{\text{stp}}$ and $\varphi(\cdot)$ denotes a projection keeping the same setting with $\psi(\cdot)$.

In the channel view, DPIC utilizes a multihead attention mechanism to generate channel parameters, enabling NIR-Net to adjust the importance of different channels globally. Specifically, the query and key components of the attention mechanism are computed as the mean value of the feature across each spatial dimension. This approach ensures that the attention mechanism captures the overall trend and context across the feature, allowing it to identify the channels that

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are responsive to objects. On the other hand, the value component of the attention mechanism is the max value of the spatial sensitive feature. The attention mechanism pinpoints the most significant features within each channel by focusing on these max values, and a sigmoid activation function is applied to constrain the output within a spectrum from 0 to 1. This attention-based channel descriptor thus represents the significance of diverse channels, with the values closer to 1 indicating higher importance. Finally, the descriptor is calculated by the following formula:

$$\boldsymbol{d} = \sigma\left(\frac{\bar{\mathbf{F}} \cdot \bar{\mathbf{F}}^{\mathrm{T}}}{\sqrt{\lambda}}\right) \cdot \tilde{\mathbf{F}} \tag{7}$$

where $\sigma(\cdot)$ represents a sigmoid function and the number of channels λ equals 128. The optimal descriptor $\boldsymbol{d}_{\mathrm{opt}}$ is generated based on the max value of intensity path feature $\tilde{\mathbf{F}}_{\mathrm{inp}}$, while the holistic descriptor $\boldsymbol{d}_{\mathrm{hol}}$ is generated based on the max value of stability path feature $\tilde{\mathbf{F}}_{\mathrm{inp}}$.

C. Noise Perception Module

Straightforward feature fusion often fails in complex scenes, making it struggle to detect challenging objects [44], [72]. As a result, the static strategy leads to lower detection accuracy when encountering obscured objects, particularly in disturbed environments such as changeable cloud occlusions. To mitigate this problem, we propose an essential NPM, which identifies object areas and adjusts to different channels. The previous study [73] defines the minimum pixel intensity as the dark channel prior, which is able to estimate the thickness of haze in a direct manner. Inspired by this, NPM attempts to employ the minimum responses of feature maps to mitigate the disturbance from cloud corruption. Here, NPM employs a minimum pooling operation to isolate background corruption and emphasize the most critical information related to objects. The operation directly on the spatial sensitive feature $\mathbf{F} \in \mathbb{R}^{H\times W\times C}$ utilized in the FCM ensures seamless alignment within the dual-path calibration mechanism. As shown in Fig. 2, the channel minimization indicates the object regions with significantly higher feature values than the surrounding environment, which distills the significant noise-irrelevant details from corrupted features. We describe the discrepancy between objects and background noise as follows:

$$v(i, j) = \min_{k}(\mathbf{F}(i, j, k)) \in \mathbb{R}^{H\times W} \tag{8}$$

where $v(i, j)$ represents the minimal response at point (i, j). The discrepancy map is then simultaneously weighted by two channel-view excitations, which are the minimal values across split features, as expressed in the following equation:

$$\omega(k) = \min_{i,j} \mathbf{U}(i, j, k) \in \mathbb{R}^{C/2}. \tag{9}$$

The split sensitive feature $\mathbf{F}_{\mathrm{stp}}$ is utilized as the input $\mathbf{U} \in \mathbb{R}^{H\times W\times C/2}$ for stability path calibration weight $\mathbf{W}_{\mathrm{stp}}$, while $\mathbf{F}_{\mathrm{inp}}$ for intensity branch $\mathbf{W}_{\mathrm{inp}}$. The calibration weight is a combination of extracted discrepancy map v and channel excitation ω, which are crucial for guiding the dynamic feature calibration, as described in the following equation:

$$\mathbf{W}(i, j, k) = \sigma(v(i, j) \times \omega(k)) \in \mathbb{R}^{H\times W\times C/2} \tag{10}$$

where $\sigma(\cdot)$ denotes a sigmoid activation function guaranteeing that the excitation value falls within the range of 0–1. Therefore, NIRNet can effectively adjust its focus to object areas and emphasize their representations across different channels using the perceived weights from corrupted features.

IV. Experimental Results and Analysis

In this section, we introduce the datasets, experimental setup, metrics, and main results. Furthermore, we conduct extensive ablation studies to validate the robustness of NIRNet against cloud corruption.

A. Dataset

1) Hazy-DIOR: We put forward a new benchmark, Hazy-DIOR, to better examine the robustness of object detection models in remote sensing images obscured by clouds or fog. In brief, we generate diverse hazy images for the dataset by incorporating simulated clouds into the raw geoscience images sourced from the testing set of the DIOR-R dataset [74], [75]. DIOR-R is a large-scale benchmark for remote sensing object detection, which consists of 23 463 images spanning 20 categories, including airplane (APL), airport (APO), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), dam (DAM), expressway service area (ESA), expressway toll station (ETS), harbor (HA), golf field (GF), ground track field (GTF), overpass (OP), ship (SH), stadium (STA), storage tank (STO), tennis court (TC), train station (TS), vehicle (VE), and windmill (WM). The total images are divided into three subsets: training set, validation set, and testing set, comprising 5862, 5863, and 11 738 images, respectively.

How to Synthesis Hazy Images? When observing discernible objects, the high-altitude clouds with low thickness typically appear in remote sensing images. These clouds usually manifest as wisps rather than distinct patterns [69], which is neglected in the previous synthesis arithmetic of hazy images [76]. To accommodate this scenario, we gather numerous simulated cloud images from Internet and crop them into 360 patches, which are intuitively categorized into three types: cirrus, cirrocumulus, and cirrostratus, as exampled in Fig. 3. In line with established techniques in computer vision [77], [78], [79], [80], [81], we synthesize diverse hazy remote sensing images based on the optical image model, a formulation commonly employed in image dehazing, underwater image restoration, and other weather-tolerant tasks. Specifically, the cloud patches are initially scaled to match the spatial resolution of DIOR-R images (800×800) and subsequently integrated into the clean remote sensing images on an elementwise basis with attenuated brightness. Mathematically, the formulation of the simulated hazy image $\mathbf{I}$ is defined as follows:

$$\mathbf{I} = \Gamma(\mathbf{R}) \cdot t + \mathbf{P} \cdot (1 - t) \tag{11}$$

where $\mathbf{R}$ denotes a remote sensing image, and its brightness is reduced to 80% using the function $\Gamma(\cdot)$. The cloud patch $\mathbf{P}$ and the hyperparameter $t \in [0.5, 0.8]$ are randomly assigned during the synthetic process. In a comparable fashion to the previous works [82], [83], we provide hazy remote sensing images with different corruption levels to comprehensively

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TABLE I
COMPARISON RESULTS ON THE HAZY-DIOR DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	mAP	rPC	APL	APO	BF	BC	BR	CH	ESA	ETS	DAM	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM
One-Stage :																						
Rot RetinaNet [11]	33.97	62.20	35.31	10.65	44.18	77.50	3.59	56.99	41.01	34.99	6.66	39.92	44.41	11.60	18.06	31.86	31.51	47.83	72.47	17.03	19.80	34.06
Rot FCOS [41]	35.51	59.43	35.00	16.65	44.15	71.20	13.51	42.51	36.51	38.12	9.43	29.01	45.69	10.62	27.11	51.97	36.20	49.70	72.30	17.39	22.47	40.58
R3Det [19]	36.21	63.90	33.72	12.74	48.30	71.42	13.32	54.95	43.15	41.69	11.79	29.76	43.82	13.97	31.30	40.73	38.04	45.21	72.22	19.85	20.81	37.40
S2ANet [18]	38.36	63.50	40.31	14.13	50.11	79.56	16.51	50.85	45.12	44.40	7.33	28.52	48.68	15.33	35.18	42.65	43.21	48.13	72.48	20.24	24.90	39.61
Two-Stage:																						
Rot Faster R-CNN [10]	36.82	61.31	35.73	11.02	44.81	77.37	12.21	55.43	44.10	41.62	13.50	30.59	48.92	20.18	28.83	35.81	39.16	44.41	72.44	25.53	16.66	37.96
RoI Transformer [16]	37.27	62.37	35.89	12.90	45.15	80.23	12.71	52.64	46.52	35.44	11.57	32.54	50.26	19.51	29.10	36.01	39.80	53.10	72.52	19.64	22.21	37.68
Oriented R-CNN [17]	38.70	61.76	35.18	16.86	45.18	79.33	17.48	52.36	48.00	41.29	17.02	32.69	50.30	20.02	32.73	44.91	39.44	52.85	72.49	21.64	16.68	37.56
NIRNet (Ours)	**40.86**	**64.32**	36.14	19.32	45.05	79.63	17.81	58.78	54.54	47.83	15.72	31.50	50.55	18.61	32.52	45.04	43.94	52.51	72.48	29.59	23.26	42.44

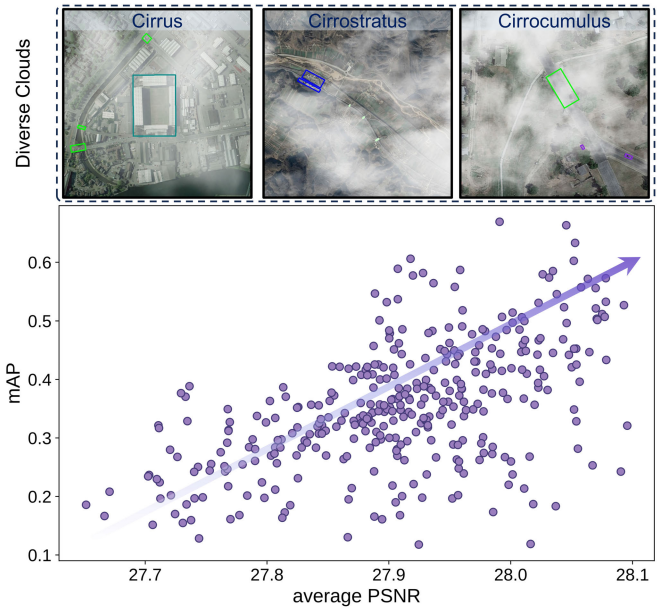


Fig. 3. Analysis of various cloud conditions. (Top) We present the synthesis results of different cloud occlusions, namely, cirrus, cirrocumulus, and cirrostratus. (Bottom) Performance degradation across different conditions: a correlation between disturbance severity and accuracy decrease. We evaluate the effect of each cloud on 300 images selected from the DIOR-R testing set. The average PSNR indicates the mean peak signal-to-noise ratio between the 300 cloud-affected conditions and their corresponding clean versions, and mAP represents the mAP of Oriented R-CNN [17] on the 300 hazy images.

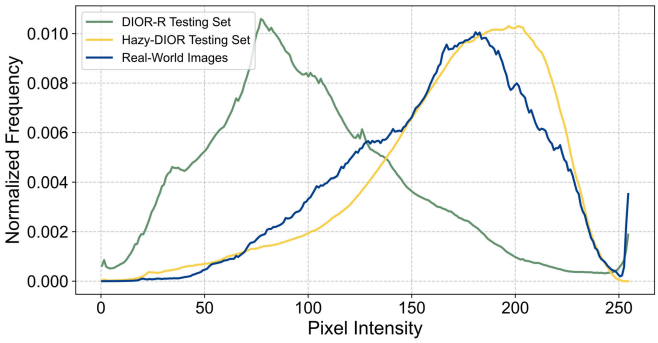


Fig. 4. Pixel intensity distribution across three datasets: Hazy-DIOR testing set, DIOR-R testing set, and 100 real-world hazy images selected from ZY-3 satellite. Both DIOR-R and Hazy-DIOR employ entire testing images.

evaluate the robustness of detectors. Nevertheless, unlike these studies that apply different corruption levels separately across the entire dataset, we repeatedly incorporate all cloud patches into clean images with different transparency hyper-parameters t. Table I summarizes the detection accuracy of mainstream aerial detectors on the Hazy-DIOR dataset, which reveals an approximately 40% performance decline across all detectors.

Aiming to validate the image quality of the constructed Hazy-DIOR dataset, we statistic the pixel intensity distribution of DIOR-R testing set, Hazy-DIOR testing set, and 100 real-world hazy images downloaded from the ZY-3 satellite. As depicted in Fig. 4, the frequency distribution of Hazy-DIOR resembles that of real-world hazy images, where the intensity frequency exhibits a prominent peak at high pixel values. This observation contrasts with clear remote sensing images, which typically present a more balanced distribution across low to mid-range intensities due to their diverse and rich color compositions. The concentration of pixel values at the higher end in Hazy-DIOR arises from the presence of extensive bright cloud regions, which occupy a large portion of the image and contribute a significant amount of high-intensity (white) pixels. Consequently, the distribution appears more condensed, indicating a reduced tonal variety caused by haze coverage.

How to Affect Detection Accuracy? For evaluating the decline in detection accuracy across different cloud corruption types, we select 300 raw images from the DIOR-R testing set. On this basis, we generate 360 additional tiny sets, each incorporating an identical cloud patch into the entire collection. We calculate the average peak single-to-noise (PSNR) between hazy images and their corresponding clean counterparts across the tiny set to quantify the perturbation from cloud damage. Furthermore, we assess the accuracy decrease per cloud patch based on the detection results of its corresponding hazy tiny set using the Oriented R-CNN [17] detector. As shown in Fig. 3, the detection accuracy degrades progressively with increasing cloud disturbance, and a severe perturbation will lead to a significant drop. It can be attributed to the fact that thicker cloud corruption types substantially reduce the similarity between the synthesized hazy images and their clean counterparts, as evidenced by a lower average PSNR. A diminished value of the average PSNR reflects the greater loss of visual information and more severe corruption, thereby complicating the task of accurate object detection. Notably, these cloud patches are randomly selected to synthesize hazy remote sensing images in random transparency values,

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which enables Hazy-DIOR to be a more comprehensive evaluation of the robustness performance in remote sensing object detection.

2) DOTA-Cloud: DOTA-Cloud [76] is a cloud noise remote sensing dataset based on the DOTA dataset [84]. It contains 2086 images and 188 282 instances with oriented bounding box annotations. The training set consists of clean images, while the testing set comprises hazy remote sensing images, covered by 15 categories: plane (PL), baseball diamond (BD), BR, GTF, small vehicle (SV), large vehicle (LV), ship (SH), TC, BC, storage tank (ST), soccer-ball field (SBF), roundabout (RA), HA, swimming pool (SP), and helicopter (HC). Image dimensions range from 800 × 800 to 4000 × 4000 pixels in the DOTA-Cloud dataset. All images are then cropped to a resolution of 1024 × 1024, with a gap of 200 pixels.

B. Experimental Setup

We conduct extensive experiments on the MMRotate platform to evaluate the proposed NIRNet. The dataset partition of Hazy-DIOR is consistent with DIOR-R. Unless otherwise specified, the mainstream backbone ResNet-50 [1] with the basic FPN [44] is adopted as the feature extractor, and Oriented R-CNN [17] is utilized as the baseline detector. For model training, we use a single RTX3090 GPU with a batch size of 2 and 12 epochs on both benchmarks. During the training phase, all models are optimized using the stochastic gradient descent algorithm with a learning rate of 0.005, setting a momentum of 0.9 and a weight decay of 0.0001. Notably, the reported GFLOPs in this article are calculated on a single RTX3090 GPU by adopting an 800 × 800 remote sensing image as input. Consistent with mainstream detectors in the remote sensing community, oriented bounding boxes are employed to evaluate the localization precision.

C. Metrics

The mean average precision (mAP) is employed to assess the accuracy of aerial detectors in hazy remote sensing scenes. The relative performance under corruption (rPC), calculated based on (12), is used to assess the robustness of detectors in coping with cloud corruption

$$\text{rPC} = \frac{\text{mAP}_{\text{hazy}}}{\text{mAP}_{\text{clean}}} \tag{12}$$

where mAP_{hazy} and $\text{mAP}_{\text{clean}}$ denote the mAPs for hazy and clean images, respectively, utilizing an identical model trained on raw images. The robustness of aerial detectors against some common corruptions is evaluated by the mean performance under corruption (mPC), formulated as follows:

$$\text{mPC} = \frac{1}{N_C}\sum_{C=1}^{N_C}\frac{1}{N_S}\sum_{1}^{N_S} P_{C,S} \tag{13}$$

where N_C is 15, representing the number of corruption types. Following [85], the number of severity levels N_C is set to 5 for each corruption, and the performance $P_{C,S}$ denotes the mAP in each condition.

D. Main Results

We compare various popular aerial detectors, including Rotated Faster R-CNN [10], RoI Transformer [16], Oriented R-CNN [17], Rotated RetinaNet [11], Rotated FCOS [41], R3Det [19], and S2ANet [18], on the Hazy-DIOR dataset and present their results in Table I. Similar to the generic remote sensing object detection task, one-stage detectors often exhibit slightly lower accuracy than two-stage models. However, the inaccurate recognition of challenging objects leads to poor accuracy, which may provide an impression of robustness against cloud corruption. In contrast, two-stage detectors attempt to perform better across most building-related categories, such as HA, GTF, and dam. These results suggest that the detector with sophisticated architecture could recognize the intricate objects blended with their ambient environment.

Among all two-stage detectors, NIRNet gets the best performance (40.86% mAP) on the Hazy-DIOR dataset without bells and whistles. In addition, NIRNet guarantees superior robustness with 64.32% rPC, outperforming all one-stage and two-stage detectors. The experimental results demonstrate that NIRNet further improves the accuracy of building-related categories (e.g., APO, STA, and TC), and we visualize its detection results in Fig. 5. Thanks to the effective feature calibration, NIRNet even enhances the detection accuracy on clean images, contributing an approximately 1% improvement. Notably, the rPC metric is affected by two essential factors, where the higher performance in clean images or the lower accuracy in hazy images probably leads to lower robustness in different datasets. Moreover, we present the performance of our NIRNet on the DOTA-Cloud dataset to validate its robustness comprehensively. As shown in Table II, NIRNet achieves 61.65% mAP and 81.27% rPC only with single-scale training, surpassing all other detectors in both metrics. These results suggest that noise incentive calibration is an effective strategy for addressing the disturbances posed by cloud corruption.

Considering the proposed NIRNet is the first detection model that concentrates on improving the robustness of aerial object detectors against cloud corruption, several related types of research in enhancing robustness are compared in this study. Importantly, to better exemplify the generalization of its essential component (DPIC), the typical one-stage detection framework (rotated FCOS [41] with the basic FPN [44]) is selected as the baseline detector for comparisons. As evidenced from Table III, we integrate instance-batch normalization [86], group normalization [87], and weight standardization [88] into FPN. The poor performance of these methods in terms of accuracy could be attributed to inadequate extraction of discriminative representations. Besides these, the influential path aggregation networks (PA-FPN [89]) and residual feature augmentation (Aug-FPN [90]) are compared. Meanwhile, the neural architecture search approach (NAS-FPN [91]), which is proven to be robust for weather corruptions in [92], also provides poor robustness even though training for 50 epochs. The quantitative results validate that DPIC is a robust feature calibration network. It produces the best detection accuracy in both Hazy-DIOR and

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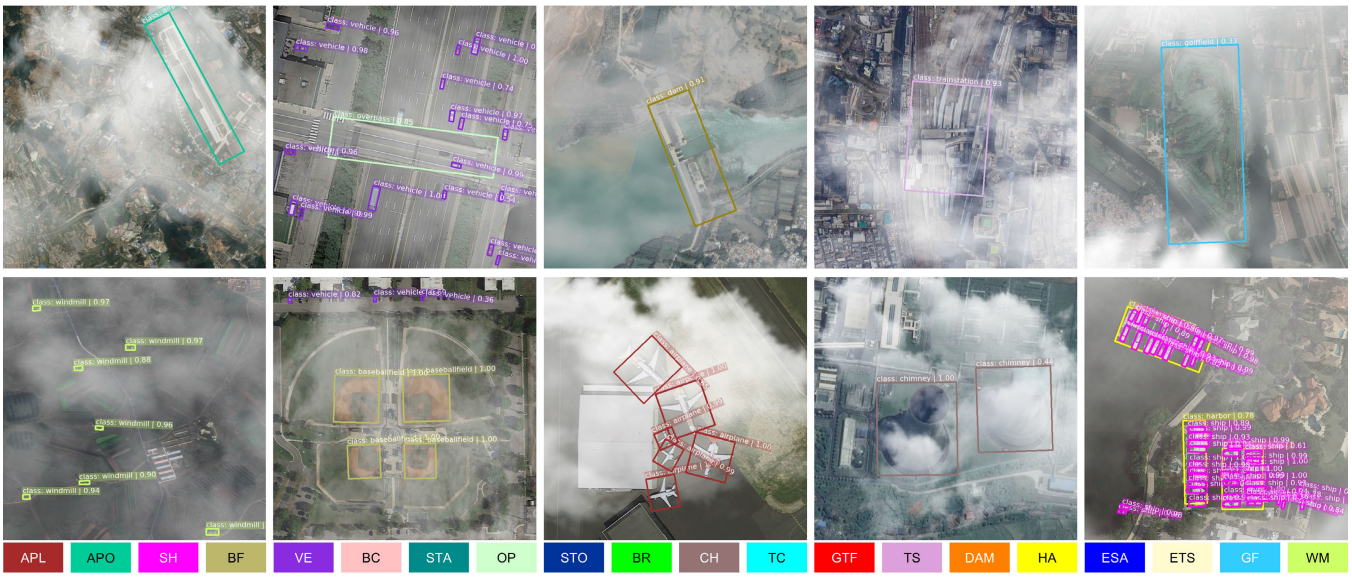


Fig. 5. Visualization results of NIRNet for Hazy-DIOR, with detection outputs shown as colored bounding boxes annotated with classification scores.

TABLE II
COMPARISON RESULTS ON THE DOTA-CLOUD DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	mAP	rPC
One-Stage:		
Rotated RetinaNet [11]	55.12	79.73
Rotated FCOS [41]	57.51	80.68
R3Det [19]	56.65	81.15
S2ANet [18]	59.26	80.22
Two-Stage:		
Rotated Faster R-CNN [10]	58.53	79.73
RoI Transformer [16]	60.03	78.90
Oriented R-CNN [17]	60.59	80.05
NIRNet (Ours)	**61.65**	**81.27**

TABLE III
COMPARISON RESULTS WITH ROBUST FEATURE FUSION METHODS ON THE HAZY-DIOR DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	mAP_{hazy}	mAP_{clean}	rPC	GFLOPs	Params	FPS
FPN [44]	35.51	59.75	59.43	126.56	31.93M	31.4
FPN-IBN [86]	36.51	59.74	61.11	126.56	31.93M	30.9
FPN-GN [87]	36.09	59.53	60.62	126.56	31.93M	32.5
FPN-GN-WS [88]	36.10	58.86	61.33	116.40	31.94M	31.9
PA-FPN [89]	36.84	59.53	61.88	130.25	34.29M	14.5
Aug-FPN [90]	34.82	59.24	58.78	343.99	33.19M	10.3
NAS-FPN† [91]	35.89	60.20	59.62	194.27	58.06M	9.1
FPN-DPIC (Ours)	**38.27**	**60.70**	**63.05**	142.34	33.12M	27.8

The baseline in this comparison is one stage detector Rotated FCOS [41], and the symbol † represents training for 50 epochs.

DIOR-R benchmarks, meanwhile achieving the highest rPC with an acceptable efficiency drop.

Except for cloud corruption, the performance of NIRNet is quantified in another DIOR-R robustness benchmark, which consists of 15 common corruption conditions in imagery and each one covers five different severity levels. Following the previous research [85], we leverage the mPC metric to evaluate the robustness against corruption, and the results are reported in Table IV. Profiting from its robust DPIC, the proposed NIRNet achieves the highest mPC across diverse corruption types with different severity levels. These global comparisons share a similar qualitative conclusion with the Hazy-DIOR benchmark, in which NIRNet achieves the best accuracy in all noise (Gaussian, shot, and impulse) and weather (snow, frost, and fog) corruption types. Remarkably, the superior performance across all weather-tolerated scenarios highlights its strong generalization capability in object detection against diverse weather conditions.

E. Ablation Studies

In this section, we conduct a series of ablation studies on the Hazy-DIOR dataset, presenting quantitative and qualitative results to validate the performance of NIRNet.

1) Generalization of DPIC: We first examine the generalization of the DPIC strategy. Specifically, we employ the incentive calibration mechanism in a diverse set of representative one-stage and two-stage object detection frameworks to investigate its effect on accuracy and robustness. As demonstrated in Table V, the incorporation of DPIC consistently contributes to the remarkable improvement across all tested models. The detectors with DPIC mechanism significantly surpass their corresponding competitors by 1.5% in mAP and 2.0% in rPC. As evidenced in Fig. 1, incorporating DPIC with Oriented R-CNN [17] notably mitigates the imprecise localization, especially in angle regression, and effectively avoids misclassification for some partially occluded instances. These results underscore the effectiveness and generalization of DPIC in generating discriminative representations to boost the accuracy and robustness of aerial detectors.

As a feature calibration approach, the generalization capability of DPIC is further examined by integrating it into mainstream feature extraction networks. Table VI

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TABLE IV

COMPARISON RESULTS ON THE DIOR-R CORRUPTION DATASET, IN WHICH THE DETECTION ACCURACY IS THE AVERAGE MAP ON FIVE SEVERITY LEVELS, AND THE MPC DENOTES THE MEAN PERFORMANCE UNDER DIVERSE CORRUPTION TYPES. THE BEST RESULT IS HIGHLIGHTED IN BOLD

Methods	mPC	Noise			Blur				Weather			Digital				
		Gaussian	Shot	Impulse	Defocus	Glass	Motion	Zoom	Snow	Frost	Fog	Brightness	Contrast	Elastic	Pixelate	JPEG'
One-Stage:																
Rot RetinaNet [11]	30.58	14.53	13.93	11.83	27.15	30.71	32.54	21.23	20.35	32.23	39.48	48.77	30.36	48.12	42.25	45.18
Rot FCOS [41]	31.98	14.32	11.45	29.34	31.68	33.57	22.35	22.85	33.21	43.52	50.22	31.59	48.27	45.20	47.45	56.44
R3Det [19]	34.10	18.06	16.36	29.84	33.32	35.88	21.45	23.66	35.06	46.22	53.00	34.88	50.71	45.20	48.90	57.06
S2ANet [18]	34.44	17.72	16.08	28.94	33.60	33.92	21.66	26.18	35.76	45.98	53.78	35.14	53.98	45.28	50.40	57.02
Two-Stage:																
Rot Faster R-CNN [10]	33.69	15.48	15.54	13.78	28.44	33.25	35.09	22.59	25.21	34.66	45.35	52.91	33.60	52.83	46.40	50.27
RoI Transformer [16]	34.26	17.58	16.53	30.08	33.46	34.86	22.62	24.26	34.24	45.80	53.33	34.54	51.82	46.52	49.92	57.00
Oriented R-CNN [17]	36.29	18.62	16.72	32.57	35.50	36.92	24.98	26.58	36.45	48.22	56.02	36.52	54.94	48.76	52.82	57.92
NIRNet (Ours)	**37.08**	18.92	17.30	33.96	38.10	37.52	23.66	27.08	37.18	49.84	56.82	37.78	55.10	49.44	53.84	58.49

TABLE V

ABLATION EXPERIMENTS ON THE GENERALIZATION FOR POPULAR AERIAL DETECTORS. NUMBERS IN BRACKETS REPRESENT THE IMPROVEMENT BY USING THE DPIC

Models	DPIC	mAP_{hazy}	mAP_{clean}	rPC
One-Stage:				
R3Det [19]	✗	36.21	56.67	63.90
	✔	38.41 (+2.20)	57.16 (+0.49)	67.20 (+3.30)
S2ANet [18]	✗	38.36	60.41	63.50
	✔	40.11 (+1.75)	60.90 (+0.49)	65.86 (+2.36)
Two-Stage:				
RoI Transformer [16]	✗	37.27	59.76	62.37
	✔	38.80 (+1.53)	60.20 (+0.45)	64.45 (+2.08)
Oriented R-CNN [17]	✗	38.70	62.66	61.76
	✔	40.86 (+2.16)	63.53 (+0.86)	64.32 (+2.56)

TABLE VI

ABLATION EXPERIMENTS ON THE GENERALIZATION FOR DIFFERENT FEATURE EXTRACTORS. NUMBERS IN BRACKETS REPRESENT THE IMPROVEMENT BY USING THE DPIC

Models	DPIC	mAP_{hazy}	mAP_{clean}	rPC
ResNet-50 [1]	✗	38.70	62.66	61.76
	✔	40.86 (+2.16)	63.53 (+0.87)	64.32 (+2.55)
LSKNet-S [67]	✗	33.37	62.96	53.00
	✔	36.45 (+3.08)	63.35 (+0.39)	57.54 (+4.54)
Swin-T [43]	✗	46.68	64.78	72.06
	✔	48.70 (+2.02)	65.10 (+0.32)	74.81 (+2.75)

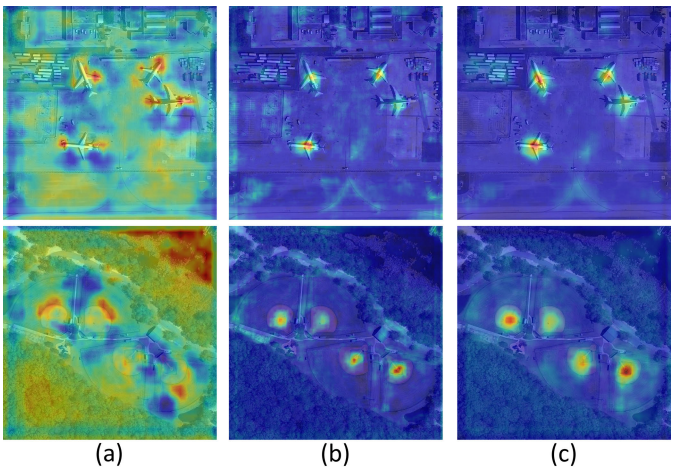


Fig. 6. Illustration of feature maps for detection heads. (a) Oriented R-CNN [17] struggles to identify objects. (b) Applying the incentive before calibration successfully localizes objects. (c) Applying the incentive after calibration improves object recognition.

reports the empirical results for several typical feature extraction backbones in remote sensing object detection, including a convolution-based backbone (deep residual networks (ResNets) [1]), a Transformer-based backbone (Swin Transformer-tiny (Swin-T) [43]), and a remote sensing backbone (large selective kernel networks (LSKNets) [67]). The comparison results demonstrate that DPIC contributes to a more significant improvement on the DIOR-R benchmark for ResNet than LSKNet, which illustrates that the dual-path calibration mechanism strengthens the conventional backbone to generate more discriminative representations. The highest enhancement on the rPC metric of LSKNet validates that the proposed noise incentive strategy could effectively mitigate the accuracy degradation from cloud occlusion. Besides convolution-based backbones, DPIC could naturally generalize to the Transformer-based backbone with considerable enhancements both in accuracy and robustness performance.

2) Individual Contribution: We perform ablation experiments to evaluate the necessity of two components in DPIC: FCM and NPM. The empirical objective is to investigate both individual and collaborative improvement of these components on detection precision and robustness under cloud corruption. As detailed in Table VII, the baseline detector excluding both components markedly diminishes detection performance, evidenced by the mAP of 38.70% and rPC of 61.76%.

Solely employing NPM in the baseline detector contributes to a considerable improvement in detection accuracy, facilitating the mAP to 39.71% and rPC to 62.25%. This enhancement proves that the NPM, tasked with perceiving the severity and suppressing the adverse effects of noise corruption, is essential in differentiating the minimal responses between relevant and irrelevant representations. Although the proposed NPM delivers a positive contribution, its capacity is somewhat restricted when functioning independently due to the corrupted feature representations being weighted. Consequently, we develop the FCM to generate discriminative feature representations for

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TABLE VII
ABLATION EXPERIMENTS FOR INDIVIDUAL CONTRIBUTION OF EACH COMPONENT IN DPIC

NPM	FCM	mAP	rPC
✗	✗	38.70	61.76
✔	✗	39.71	62.25
✗	✔	40.53	63.34
✔	✔	40.86	64.32

TABLE VIII
ABLATION EXPERIMENTS FOR INCENTIVE POSITION

Incentive Position	mAP	rPC
Before Calibration	40.74	64.31
After Calibration	40.86	64.32

TABLE IX
ABLATION EXPERIMENTS FOR PATH SCHEME

Calibration Strategy	mAP	rPC
Baseline	38.70	61.76
Double Intensity-Path	40.67 (+1.97)	63.98 (+2.22)
Double Stability-Path	40.34 (+1.64)	63.61 (+1.85)
Dual-Path	40.86 (+2.16)	64.32 (+2.56)

TABLE X
ABLATION EXPERIMENTS FOR CHANNEL DESCRIPTOR

Descriptor Mechanism	mAP	rPC
Spatial-wise Average	40.22	63.44
Spatial-wise Max	40.40	63.26
Attention Mechanism	40.86	64.32

potential objects. The employment of FCM alone yields a more pronounced increase in detection performance, with the mAP rising to 40.53% and rPC to 63.34%. The significant gain in both metrics proves that FCM is highly effective in generating prominent features, thereby improving the detection accuracy.

Integrating FCM and NPM into the baseline detector results in exceptionally remarkable improvements in detection performance. Significantly, this combination contributes to the maximum enhancement, achieving 40.86% mAP and 64.32% rPC. Therefore, the synergy between these two pivotal components appears to be particularly potent. The FCM ensures prominent feature extraction by prioritizing attention on the optimal and holistic feature representations, and the NPM further refines them by dynamically adjusting the weights according to the perceived discrepancy between objects and noise. Generally, this combination leverages the strengths of both components, resulting in a more accurate and robust detection. It is worth emphasizing that these promising results are achieved without utilizing any hazy remote sensing images during training. It highlights the inherent robustness of NIRNet, which benefits from the noise incentive calibration and discriminative enhancement provided by the DPIC module. Unlike large-scale synthetic pretraining methods [93], [94], [95] that rely on extensive labeled data from simulated environments, our approach focuses on the specific challenge (cloud occlusion) and introduces a lightweight yet effective solution. As shown in Table IV, NIRNet demonstrates consistent performance improvements under diverse corruption types and severity levels, further validating its generalization capacity in practical remote sensing scenarios.

3) Incentive Position: Our investigation primarily delves into how the incentive position of NPM determines detection accuracy and robustness. Therefore, we examine two possibilities: positioning the NPM preceding or following the dual-path feature extraction. The latter pattern is demonstrated in Fig. 2, and the former attaches NPM to the depthwise convolution layer. As depicted in Table VIII, the comparison results reveal that positioning the NPM after prominent feature extraction contributes to a slight but noticeable improvement in detection accuracy. Following the application of positioning NPM after feature calibration, mAP is enhanced from 40.74% to 40.86%. As illustrated in Fig. 6, the results indicate that implementing the NPM after calibration enhances valid representations and feature refinement, ultimately boosting detection precision.

4) Path Scheme: Our experiments confirm that employing the two parallel paths with different focuses is a good choice for efficient feature calibration. In detail, we conduct a comparative analysis encompassing three variations of double-path feature calibration strategies: double stability-path calibration, double intensity-path calibration, and dual-path calibration. It is crucial to recognize that the double stability-path and double intensity-path strategies both involve the repetitive application of stability-path or intensity-path calibration, respectively. The primary distinction between these two paths lies in the method of prominent feature extraction.

The results of our experiments, as shown in Table IX, reveal that double stability-path calibration performs comparably to double intensity-path calibration, with both approaches leading to significant performance improvements over the baseline. In particular, the dual-path strategy, which combines the stability-path and the intensity-path calibration, exhibits the best results. This approach contributes a 2.16% improvement in mAP and a 2.56% gain in rPC compared to the baseline model (Oriented R-CNN [17]). These findings demonstrate the effectiveness of dual-path strategies in extracting prominent representations for challenging objects in hazy remote sensing images.

5) Channel Descriptor: The channel descriptor we developed, which closely combines global and local spatial contexts to prioritize prominent attention, exhibits a distinct superiority compared to alternative descriptor methods. In particular, we compare the proposed attention-based descriptor against an average channel descriptor (spatialwise global average pooling) and a max channel descriptor (spatialwise global max pooling) [96]. As listed in Table X, the attention-based channel descriptor markedly surpasses its counterparts, evidenced by a receivable rise in mAP to 40.86% and rPC to 64.32%.

This superior performance highlights the effectiveness of the attention mechanism in improving the capability of DPIC to capture the most relevant contexts, thereby providing discriminative features for detection heads. In contrast, the spatialwise

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pooling mechanisms, which either average out or maximize spatial information, fall short of capturing the intricate spatial relationships necessary for high-performance detection. Consequently, the global context of the attention-based approach contributes to a more stable and reliable detection performance against diverse corruption.

V. Conclusion

In this article, we design an NIRNet for remote sensing object detection against cloud corruption without relying on hazy images for training. In particular, the NPM identifies the areas of potential objects and weights them in different channels to generate pointwise excitations. Building upon this, the proposed DPIC mechanism further leverages these excitations to emphasize the discriminative features. Additionally, constrained by inadequate dataset coverage to evaluate the robustness of aerial detectors against diverse cloud corruptions, we construct and assess the Hazy-DIOR dataset. Comprehensive experimental results demonstrate that NIRNet effectively mitigates misclassification and imprecise regression, achieving robust detection under cloud corruption.

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Peng Zhang received the B.S. degree from Northwestern Polytechnical University, Xi'an, China, in 2023, where he is currently pursuing the Ph.D. degree, under the supervision of Prof. Gong Cheng.

His research interests are object detection and remote sensing image understanding.



Gong Cheng (Senior Member, IEEE) received the B.S. degree from Xidian University, Xi'an, China, in 2007, and the M.S. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, in 2010 and 2013, respectively.

He is currently a Professor with Northwestern Polytechnical University. His main research interests are computer vision and pattern recognition.

Dr. Cheng is an Associate Editor of *IEEE Geoscience and Remote Sensing Magazine* and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Chunbo Lang received the B.Eng. degree from Northeast Forestry University, Harbin, China, in 2020. He is currently pursuing the Ph.D. degree with Northwestern Polytechnical University, Xi'an, China, under the supervision of Prof. Gong Cheng.

His research interests are few-shot learning, segmentation, and remote sensing image understanding.



Xingxing Xie received the B.S. degree from Inner Mongolia University, Huhhot, China, in 2015, and the M.S. degree from Northwestern Polytechnical University, Xi'an, China, in 2018, where he is currently pursuing the Ph.D. degree.

His main research interests are computer vision and pattern recognition.



Junwei Han (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees in pattern recognition and intelligent systems from Northwestern Polytechnical University, Xi'an, China, in 1999, 2001, and 2003, respectively.

He was a Research Fellow with Nanyang Technological University, Singapore, The Chinese University of Hong Kong, Hong Kong, the Dublin City University, Dublin, Ireland, and the University of Dundee, Dundee, U.K., from 2003 to 2010. He is currently a Professor with Northwestern Polytechnical University. His research interests include computer vision and brain imaging analysis.

Dr. Han is an Associate Editor of IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE and IEEE TRANSACTIONS ON MULTIMEDIA.

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